Physics Note

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1 *Electric Field*

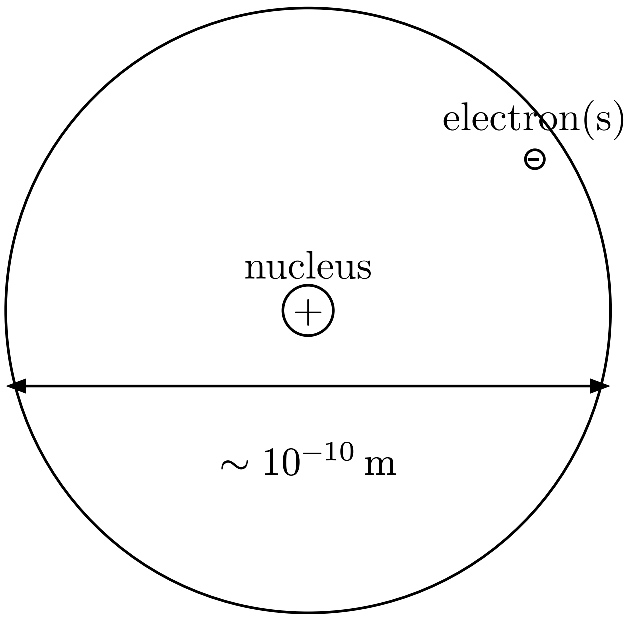
1.1 Electrostatic Force

– **electrification** - A process getting charge
– **static electoricity** - static charge
– **electoric charge** -
– **point charge** - charge which can be ignored the size

Electrostatic force is an interraction between charged particles.

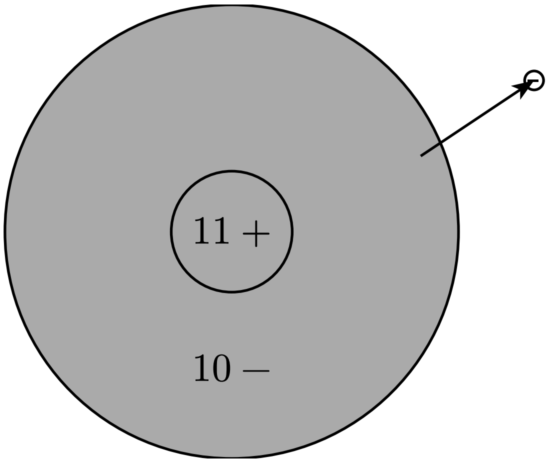
$$|\mathbb{F}| = k\frac{q_1 q_2}{r^2}$$

1.2 Structure of Atom



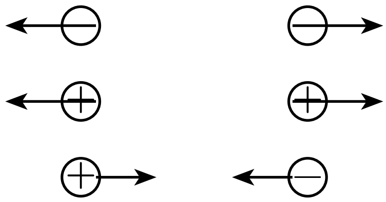
– **elementary charge** $1.6 \cdot 10^{19}$ C

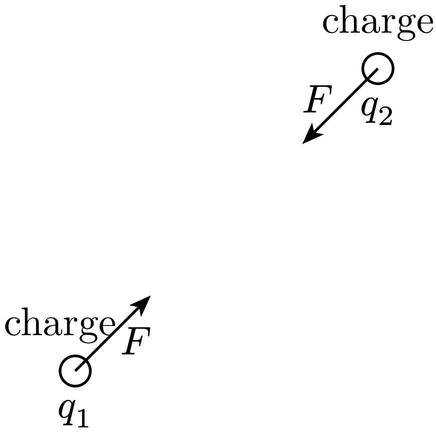
1.3 Mechanism of Electrification



Total amount of charge is conserved.

– **Coulomd's law**





$$F = k\frac{q_1 q_2}{r^2}$$

k depends on type of surrounding substance.

In bacum, $k \approx 9.0 \cdot 10^9\,\mathrm{N\,m^2\,C^{-2}}$

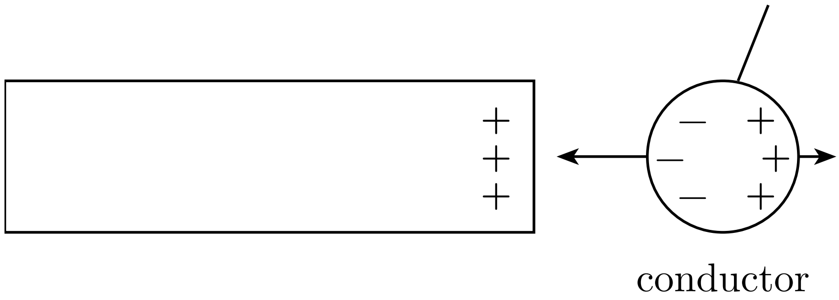
1.4 Electrostatic Indcuction

– **Conductor** - Substances which can conduct electricity.

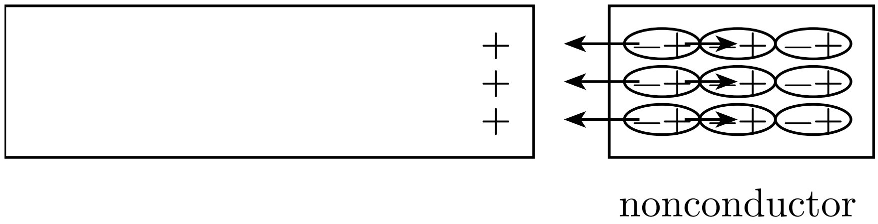
ex) metal, carbon

- **Nonconductor** - Substances which cannot conduct electricity.
- **Semiconductor** - Substances which have middle electric resistance between conductor and nonconductor.

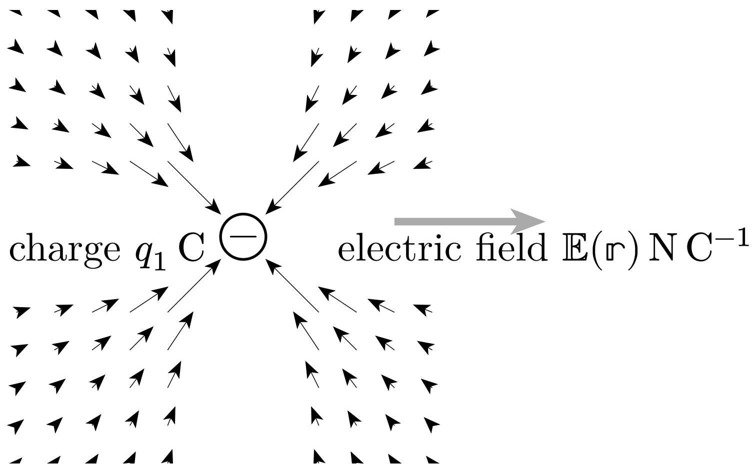
- **indcuction**



- **Dielectric Polarization**



1.5 Electric Field



Electric field is vector field.

Let 1 C charge be a test charge at position $\mathbb{r}$, $\mathbb{E}(\mathbb{r})$ is the electric force which the test charge receive.

$$\mathbb{E}(\mathbb{r}) = kQ\frac{\mathbb{r}}{|\mathbb{r}|^3}$$

If there are multiple charges, the electric field is equals to sum of each electric fields made by these charges.

1.6 Electric line of force

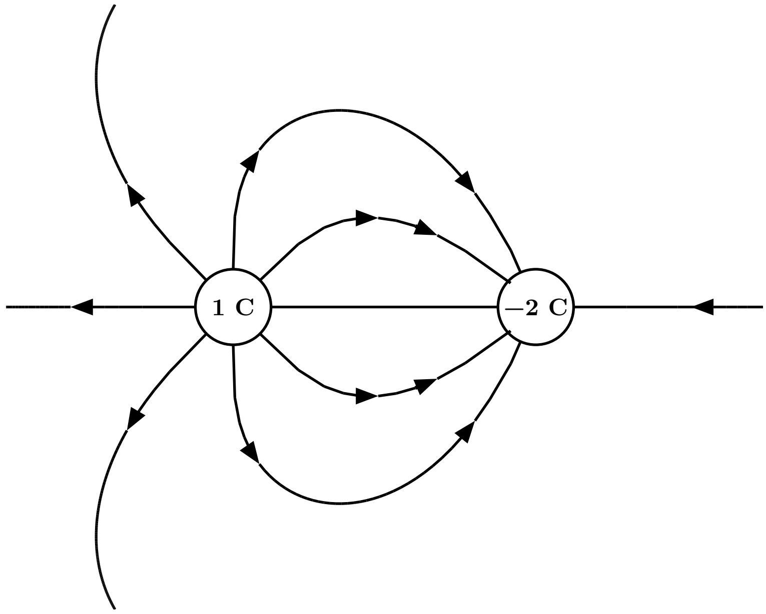


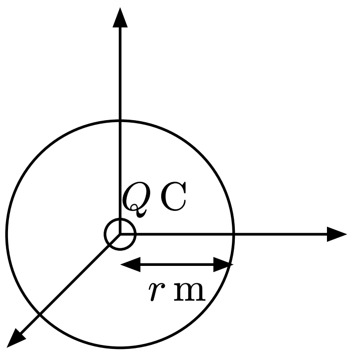
Figure 1: simulator

(it was difficult to draw the figure...)

– **the direction of tangent of line is the same as the direction of the field's vector.**

– **the line appears with plus charge and disappear with minus charge.**

– $\boldsymbol{E}$ **lines drawn per** $\mathbf{1\,m^2}$ **, as strength of electric field is** $\boldsymbol{E}\,\mathbf{N\,C^{-1}}$**.**



Let N be number of lines,Q amount of charge.

$N = E \cdot 4\pi r^2 = k\frac{Q}{r^2} \cdot 4\pi r^2$

Thus,

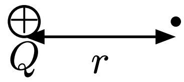
$$N = 4\pi k Q$$

1.7 Electric Potential

Let U potential energy of charge which have q, V electric potential,

$$V := \frac{U}{q}$$

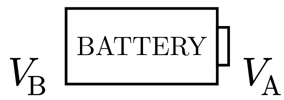
Thus, V is potential energy per unit charge.



Let infinity be reference level of potential,

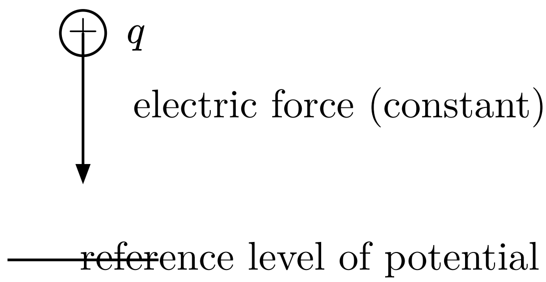
$V(r) = \int_r^\infty dr \cdot k\frac{Q}{r^2} = kQ\left[-r^{-1}\right]_r^\infty = k\frac{Q}{r}$

$$\boldsymbol{V(r) = k\frac{Q}{r}}$$



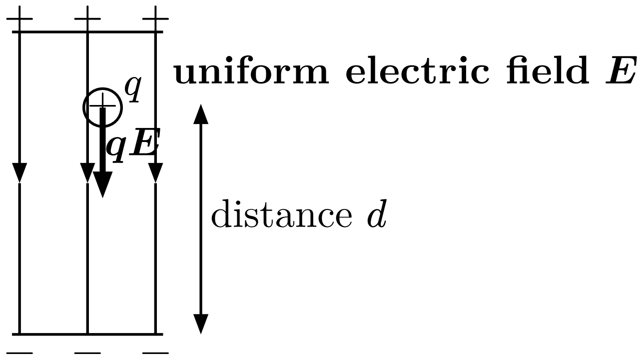
$V_{\mathrm{AB}} = V_A - V_B$ (reference is B)

V_{AB} is also called **voltage**.



$$W_{\mathrm{AB}} = q(V_A - V_B)$$

1.8 Relative of Field and Potential



$$qEd = qV$$
$$\Leftrightarrow V = Ed$$

electric potential is proportional to energy, so it's superposition is calculated by sum.

1.9 Equipotential Surface

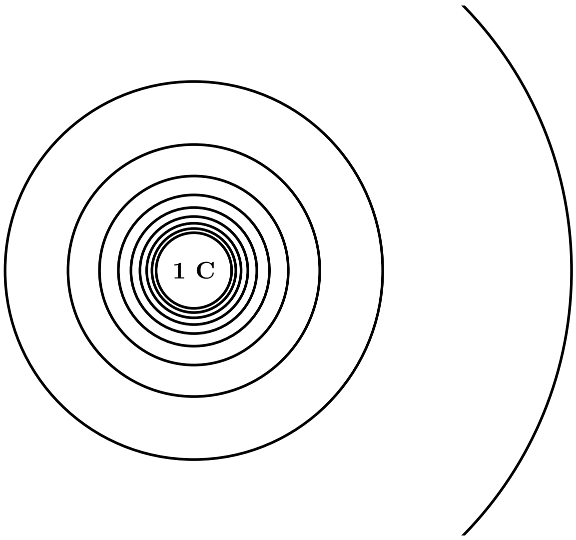


Figure 2: simulator

– **Equiipotential Surface** - A surface which was made by joyning points that have equal electric potential.

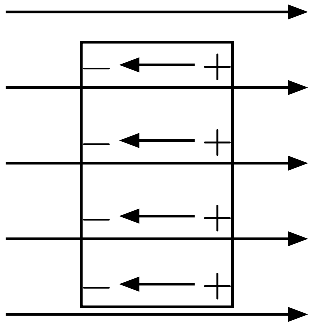
equipotential surface is perpendicular to electric line of force.

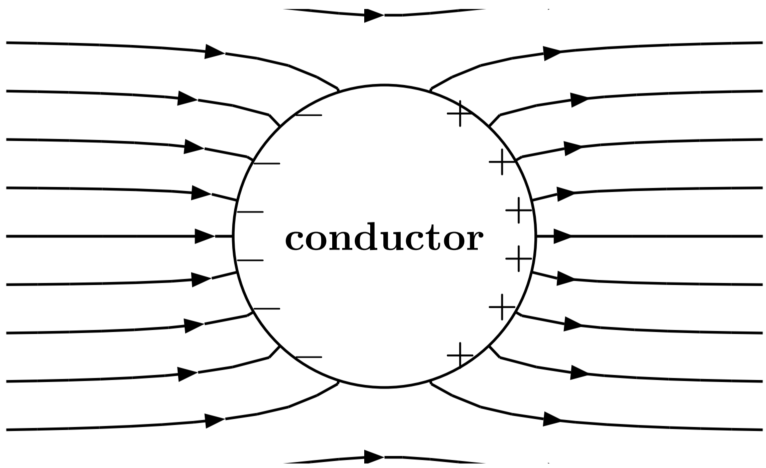
(if test charge move with a direction perpendicular to electric force , work is $W = \mathbb{F} \cdot \Delta \mathbb{x} \Leftrightarrow$ potential is same constantly.)

$$\frac{1}{2}mv^2 + qV = \text{const.}$$

1.10 Substance and Electric Field

1.10.a Conductor





In conductor, the electric field is $\mathbb{0}$. (charges contiune moving while field is not $\mathbb{0}$,)

It means that electric potential is the same in metal.

– **Electric field is perpendicular to surface of metal.** (metal distorts surrounding electric field)

– **Electric charges do not appear in internal of metal.** (on surface only)

(why? is not impossible to make field $\mathbb{0}$ with it having charges in metal? そうかな…そうかも…引力と斥力があるから自動的に表面にでできるかね。べつに全電荷が打ち消しに貢献する必要はなく、正負が surface に交互に並んでいればいい？)

– **Electric Shielding** - In cavities in conductor, electric field is $\sim \mathbb{0}$. It means field in the cavity is not affected by the outside field.

(これマジでなんで ??)

1.10.b Nonconductor

In nonconductor, the electric field is smaller thant the ouside field.

(due to the electric field generated by dielectric polarization)

1.11 Capacitor

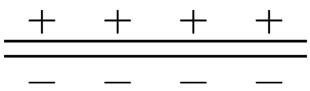


Figure 3: parallel-plate capacitor

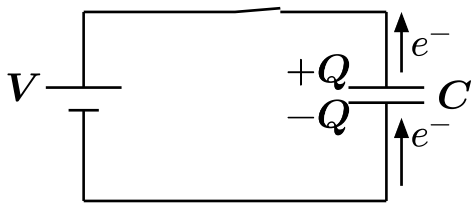


Figure 4: charging

$$\boldsymbol{C} := \frac{\boldsymbol{V}}{\boldsymbol{Q}}$$